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Hericium erinaceus enhances neurotrophic factors and prevents cochlear cell apoptosis in senescence accelerated mice

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ABSTRACT

This study aimed to investigate the effects of *Hericium erinaceus* (HE) on auditory function. Twenty-four 9-month-old male senescence-accelerated prone 8 (SAMP8) mice were randomly and equally assigned to the control group (group A) and two groups supplemented with 215.25 (group B) or 430.5 mg/kg BW of HE (group C) for 12 weeks. Compared with group A, group C had significantly smaller threshold shifts by auditory brainstem responses at the end of study. HE increased nerve growth factor (NGF) expressions in the spiral ganglion neurons (SGNs) and the temporal lobes, and also prevented outer hair cells (OHCs) loss in the middle turn of cochlea. In addition, group C had significantly lower apoptosis-inducing factor and poly (ADP-ribose) polymerase-1 expressions in the middle turn of cochlea, brainstem and temporal lobes. In conclusions, HE could prevent hearing degeneration possibly by increasing neurotrophic expression, preventing OHCs loss, and reducing the caspase-independent apoptosis signalings.

1. Introduction

With the increase of elderly population, age-related hearing impairment (ARHI) is becoming more prevalent and has great impact on the quality of daily life. The mechanisms of ARHI involved the decreased protein levels of brain-derived neurotrophic factor (BDNF) in auditory nerves ((Rüttiger et al., 2007)), pathologies and/or cell deaths of sensory cells along the auditory system (Altschuler et al., 2015), especially in the Organ of Corti and spiral ganglion neurons (SGNs) of the cochlea (Fu, Ding, Wei, Jiang, & Salvi, 2013). Moreover, increased reactive oxygen species (ROS) level was related to the severity of ARHI in adult subjects (Hwang, Chen, Hsu, Yang, & Liu, 2012). Diet-induced obesity increased hypoxia and inflammatory responses, lowered the SGNs density, and aggravated hearing degeneration by activating both caspase-dependent and -independent apoptosis, and autophagic signaling pathways in the cochlea of CD/1 mice (Hwang, Hsu, Yu, Liu, & Yang, 2013; Hwang, 2017).

Neurotrophic factors, such as nerve growth factor (NGF) and BDNF, are essential for maintaining neuronal function (Obara & Nakahata, 2002). NGF enhanced SGNs survival under gentamicin treatment (Kimura, Nishizaki, Orita, & Masuda, 1999). NGF was also involved in post-injury repair and regeneration of the basilar papilla in the inner ear following overexposure to noise (Sliwinska-Kowalska, Rzadzinska, Rajkowska, & Malczyk, 2005). In addition, BDNF is required for the survival of vestibular ganglion and type II SGNs in the inner ear (Schimmang et al., 2003). Therefore, it might be expected to see the benefits of neurotrophic factor-like substances or their inducers on ARHI.

Mushrooms may play an important role in the prevention of many age-associated neurological dysfunctions, including Alzheimer's and Parkinson's diseases (PD) (Phan, David, Naidu, Wong, & Sabaratnam, 2015). *Hericium erinaceus* (HE), a mushroom belonging to the family *Hericiaceae*, possess antioxidative, neuroprotective and immune-modulating properties (Zhang et al., 2012). HE has been demonstrated to

Abbreviations: ARHI, age-related hearing impairment; SAMP8, senescence accelerated mouse prone 8; NGF, nerve growth factor; BDNF, brain-derived neurotrophic factor; SGNs, spiral ganglion neurons; OHCs, outer hair cells; AIF, apoptosis-inducing factor; PARP-1, poly (ADP-ribose) polymerase-1; IHCs, inner hair cells; ROS, reactive oxygen species; JNK, c-Jun N-terminal kinase; ABRs, Auditory brainstem responses; TrkA, tyrosine kinase receptor A

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stimulate NGF mRNA expression in ddY mice, and protein synthesis via activating the c-Jun N-terminal kinase (JNK) pathway in 1321N1 cells (Mori et al., 2008). Among the active compounds of HE, the neuroprotective properties of *erinacine* A have been confirmed. *Erinacine* A has demonstrated increases NGF and BDNF expressions in the central nervous systems of rats (Shimbo, Kawagishi, & Yokogoshi, 2005). In a model of global ischemic stroke, *erinacine* A improved the cell survival and lessened the proinflammatory factor expressions by suppressing the reactive nitrogen species, inducible nitric oxide synthase (iNOS), p38 MAPK and homologous protein (Lee et al., 2014). E*rinacine A*-enriched HE mycelia reduced the dopaminergic lesions and oxidative stress in the stratum and substantia nigra in PD model (Kuo et al., 2016), and also improved behavioral deficits by increasing the NGF/NGF precursor ratio, and promoting the neurogenesis in the cortical and hippocampal regions of APPswe/PS1dE9 transgenic mice (Tzeng et al., 2016). However, the effects of HE on ARHI are still unclear.

The caspase-dependent apoptosis signal pathway had been reported in the cochlea of SAMP8 mice (Menardo et al., 2012). Meanwhile, NGF could prevent caspase-dependent apoptosis via activating JNK and its downstream p75/tyrosine kinase receptor A (Trk) (Mori et al., 2008). However, the axis of HE-NGF-caspase-independent apoptosis signaling pathway was never reported in SAMP8 mice. Thus, the present study aimed to investigate this issue on aged SAMP8 mice.

2. Materials and methods

2.1. Preparation of HE

The HE used in this study was supplied by Biotechnology Center, Grape King Bio Ltd. (Taoyuan, Taiwan). In brief, it was prepared as follows: HE (BCRC 35669) purchased from Bioresources Collection and Research Center (BCRC) in Food Industry Research and Development Institute (Hsinchu, Taiwan) was maintained on potato dextrose agar at 26 °C for 15 days. After incubation, a mycelial agar block (1 cm^3) was removed, transferred to a 2-L Erlenmeyer flask containing 1.3 L synthetic medium (composed of 0.25% yeast extract, 4.5% glucose, 0.5% soybean powder, 0.25% peptone and 0.05% $MgSO_4$, adjusted to pH 4.5) and incubated for 5 days at 26 °C on a rotary shaker (120 rpm). The fermentation process was then scaled up from a 2-L shake flask to 500-L and 20-ton fermenters for 5 days and 12 days, respectively. At the end of the fermentation process, the mycelia were then harvested, lyophilized, grounded into powder, and stored in a desiccator at room temperature. The dry sample was then analyzed by high-performance liquid chromatography (HPLC) to evaluate the active erinacine A content in the mycelia based on its dry weight. The HE freeze-dried mycelium powder containing 5 mg erinacine A per g (Li et al., 2019). Moreover, the percentages of carbohydrate, protein and fat in HE mycelium powder were 59.5%, 22.8% and 3.5%, respectively.

2.2. Animals

SAMP8 mice were acquired from the Council for Senescence-Accelerated Mouse (SAM) Research, Japan, and were maintained through inbreeding in the standard animal room at Providence University. According to our pilot study, the hearing of SAMP8 mice degenerated since about 6 months of age (Supplement Fig. 1). In this study, Twenty-four 9 months old male SAMP8 mice were randomly divided into three groups (n = 8/group): group A was fed an AIN 93-M basal diet, while groups B and C were fed an AIN 93-M basal diet supplemented with 215.25 mg/kg BW/day or 430.5 mg/kg BW/day of HE , respectively, for 12 weeks. The doses of HE were decided according to the results of Li et al. (2019). Their results indicated that supplementation with 108, 215, and 431 mg/kg BW/day HE reduced the lipid oxidation, enhanced the endogenous antioxidants enzymes, and extended the lifespan in SAMP8 mice. In this study, we used the same HE as our material. Thus, we refer their results and supplemented HE with the doses of 215, and 431 mg/kg BW/day to investigate the effect of HE on the auditory function on aged SAMP8 mice. The mice were housed about 5 per cage under controlled environmental conditions (22 ± 2 °C, 65 ± 5% relative humidity, 0700–1900 h lighting period). Animals were allowed free access to drinking water and diet for 12 weeks. The food intake of the mice was recorded every day, and the mice were weighed weekly. The study protocol was approved by the Institutional Animal Care and Use Committee of Providence University, Taichung, Taiwan. (20140611-P01).

2.3. Evaluation of auditory brainstem responses (ABRs)

ABRs (Intelligent Hearing Systems, Miami, FL) were measured before the start (9 months of age) and at the end of the study (12 months of age) under general anesthesia with an intraperitoneal injection of sodium pentobarbital (65 mg/kg). Subdermal needles were used as electrodes for recording. The active electrode was inserted at the vertex; the reference electrode was ventrolateral to the left ear; and the ground electrode to the low back above the tail. Click sounds, which reflected thresholds around 4-kHz, and 8-kHz and 16-kHz tone bursts were delivered sequentially to the left ear through earphones (Telephonics Corp, Farmingdale, NY). ABR thresholds were obtained by reducing the stimulus intensity in 5-decibel sound pressure level (dB SPL) intervals and increasing the stimulus intensity in 3-dB SPL intervals to identify the lowest intensity at which ABR waves I-V were detected by one well-trained audiologist who was blinded to the groups (Chan, Cheng, & Huang, 2017). Shifts of ABR thresholds, which were defined as the increase of thresholds during 12 weeks, were recorded for each mice in three groups.

2.4. Histological and immunohistochemical staining of the cochlea

At the end of the study, the mice were sacrificed by decapitation under general anesthesia with an intraperitoneal injection of pentobarbital (65 mg/kg). Animals were perfused via an intracardiac injection of 2.5% glutaraldehyde and 4% paraformaldehyde in phosphate buffered saline (PBS: NaCl, 150 mM; KH_2PO_4, 2 mM; Na_2HPO_4 $2H_2O$, 8 mM, pH 7.4) before being euthanized. For each mouse, the temporal bones were removed, and the cochlea was perfused with the above fixative through the oval and round windows to an outlet within the apex. After 24-hr post-fixation in the same fixative at 4 °C, the temporal bones were decalcified in 10% EDTA solution, 4% paraformaldehyde, and 2.5% glutaraldehyde in PBS (pH 7.4) for 1 week at room temperature. The temporal bones were then rinsed in PBS, dehydrated through a graded series of alcohol and xylene, and embedded in paraffin.

For histologic staining, embedded cochleae were sectioned at 5-μm thickness at the mid-modiolar level. The cochlear structures were observed with hematoxylin and eosin (H & E) staining using a light microscope. The cell density of neurons in SG, within a central region of 50 × 50 μm^2 at the middle turn of cochlea, was measured under ×200 magnification. The numbers of SGN were counted by the presence of a stained nucleus. The morphology and numbers of outer hair cells (OHCs) and inner hair cells (IHCs) in the Organ of Corti of cochlea were observed under ×200 magnification.

For immunohistochemical staining, the sections were heat-treated at 95 °C in 10 mM sodium citrate buffer for 20 min, incubated in 3% H_2O_2 for 10 min at room temperature, and blocked in 5% skim milk to prevent non-specific labeling. To investigate the HE on the expressions of neurotrophic factors and caspase-independent apoptosis signaling pathway. We selected NGF, BDNF, apoptosis inducing factor (AIF) and poly (ADP-ribose) polymerase-1 (PARP-1) as the stained target proteins. The specimens were then incubated 1 h at room temperature with one of the following primary antibodies: anti-NGF (1:200, catalog #ab6199, Abcam, UK) and anti-BDNF (1:200, catalog #ab226843, Abcam, UK), rabbit anti-AIF (1:1000, catalog #PB9366, Boster, CA) and

rabbit monoclonal anti-PARP-1 (1:500, catalog #PB9309, Boster, CA), were obtained from Cell Signaling Technology (TAIGEN, TPE, R.O.C.).

Slides were washed in PBS three times, exposed to a secondary antibody for 30 min at room temperature, and stained using the DAB staining kit (R&D systems, Inc., U.S.A.) according to the manufacturer's protocol. Slides were dehydrated, mounted, and observed through a light microscope. The images were acquired using a CCD camera (Dage-MTI Inc, Michigan City, Ind) connected to a personal computer and analyzed using image analysis software (Image-Pro Plus, version 6.0; Media Cybernetics, Silver Springs, MD).

The above features were observed blindly in two adjacent cochlear sections at 10-slides apart (equals to 50 μm apart) at the same mid-modiolar level. All data were acquired from two cochleae and averaged for each mouse by one well-trained researcher. Stained regions (SGN, OHCs, and IHCs) were quantified as the percentage of the entire target area by "presence or absence method" under ×200 magnification.

2.5. Western blotting

The brainstem and temporal tissues of SAMP8 mice were dissected cut into small pieces (50–60 mg), homogenized with an appropriated volume of ice-cold lysis buffer with protease/phosphatase inhibitor cocktail, and incubated in the ice for 10 min. Cytoplasmic extraction reagent was added to the tube and centrifuged for 5 min (~16,000*g*). The supernatant was immediately transferred to a clean tube (cytoplasmic extract), and added the nuclear extraction reagent to the insoluble fraction. The tube was placed on ice and continued vortexing for 15 s every 10 min, for a total of 40 min, and centrifuged for 10 min (~16,000*g*). The supernatant was immediately transferred to another clean tube (nuclear extract). Total protein in the supernatant was measured by the BCA method (Bio-Rad, Hercules, CA). The equal amount of protein (20–50 μg) was separated through sodium dodecyl sulfate-polyacrylamide gel electrophoresis and transferred onto nitrocellulose membranes. The membranes were washed with TBST buffer (10 mM Tris-HCl, 150 mM NaCl; pH 7.5) supplemented with 0.1% Tween 20 and then blocked with TBST containing 5% nonfat dried milk. The membranes were incubated overnight with primary antibodies. After being washed three times with TBST, the membranes were exposed to secondary antibodies coupled to horseradish peroxidase for 2 h at room temperature. The membranes were washed three times with TBST at room temperature. Immunoreactivity was detected using ECL reagents. Anti-NGF (1:3000, catalog #ab6199, Abcam, UK), anti-BDNF (1:1000, catalog #ab226843, Abcam, UK), anti-beta actin (1:1000, catalog#4970, Cell Signaling Technology, USA), anti-α tubulin (1:50,000, catalog#04-1117, Millipore, USA), anti-AIF (1:10,000, catalog #PB9366, Boster, CA), anti-PARP1 (1:10,000, catalog #PB9309, Boster, CA), and anti-Lamin A (1:1000, catalog#GTX101127, Gene Tex, USA) antibodies were obtained from Cell Signaling Technology (TAIGEN, TPE, ROC). Densitometric analysis of the data was performed using FusionCapt Advance Camera and FusionCapt Advance analyzer ver. 16.07 software.

2.6. Statistical analysis

One-way ANOVA with Post-hoc Bonferroni correction was used to compare the differences of all tested items between three groups using the STATA 10 software. P values of < 0.05 were considered significant.

3. Results

3.1. Shifts of ABR thresholds

Table 1 showed the shifts of ABR thresholds after fed with an AIN 93-M basal diet (group A), an AIN 93-M basal diet supplemented with 215.25 mg/kg BW/day (groups B) or 430.5 mg/kg BW/day of HE (group C), respectively, for 12 weeks. Shifts of ABR thresholds were defined as the increase of thresholds during 12 weeks. Compared with group A (12.1 ± 3.9 dB SPL), group C (2.9 ± 2.7 dB SPL), but not group B (9.3 ± 3.5 dB SPL), had significantly smaller ABR threshold shifts by click sound stimulation at the end of this study. In addition, ABR threshold shifts were significantly lower in group C than group B by click sound stimulation ($p = 0.0002$; Post-hoc analysis, $p = 0.398$ for group B versus A, $p < 0.001$ for group C versus A, and $p = 0.007$ for group C versus B, respectively). However, the ABR threshold shifts were not significantly different among the three groups by 8-kHz or 16-kHz tone burst stimulation. In addition, the food intakes did not differ among groups A, B and C, which were 6.46 ± 0.12 gm/day, 6.69 ± 0.20 gm/day and 6.78 ± 0.21gm/day, respectively.

Table 1
Shifts of ABR thresholds in three groups.

	Group A	Group B	Group C
Click sound	12.1 ± 3.9	9.3 ± 3.5	2.9 ± 2.7*,#
8 kHz tone burst	17.9 ± 9.1	10.7 ± 6.7	8.6 ± 8.5
16 kHz tone burst	7.1 ± 6.4	4.3 ± 4.5	5.0 ± 5.0

Shift of ABR threshold is defined as the increase of threshold during 12 weeks. The data of ABR threshold shifts were shown in decibel sound pressure level (dB SPL) and expressed as mean ± standard deviation. (n = 8/group) *Significantly different with group A at $p < 0.05$. #Significantly different with group B at $p < 0.05$. ABR: auditory brainstem response.

3.2. Expressions of neurotrophic factors

Fig. 1 showed the immunohistochemistric results for NGF at the SGN in the middle turn of the cochlea. Compared to group A (0.33 ± 0.15%), groups B (0.85 ± 0.11%) and C (0.77 ± 0.17%) had increased percentage of NGF staining in the SGNs ($p = 0.0002$; Post-hoc analysis, $p < 0.001$ and $p = 0.001$, respectively). However, the BDNF expressions did not differ among three groups in the SGNs (data was not shown; $p = 0.1439$).

Fig. 2 showed the ratio of mature NGF versus β-actin by Western blotting in three groups at the end of study. Compared to group A (0.36 ± 0.08%), groups B (0.52 ± 0.16%) and C (0.53 ± 0.11%) had higher expressions of mature NGF in the brainstem with a borderline significance ($p = 0.0388$; Post-hoc analysis, $p = 0.086$ and

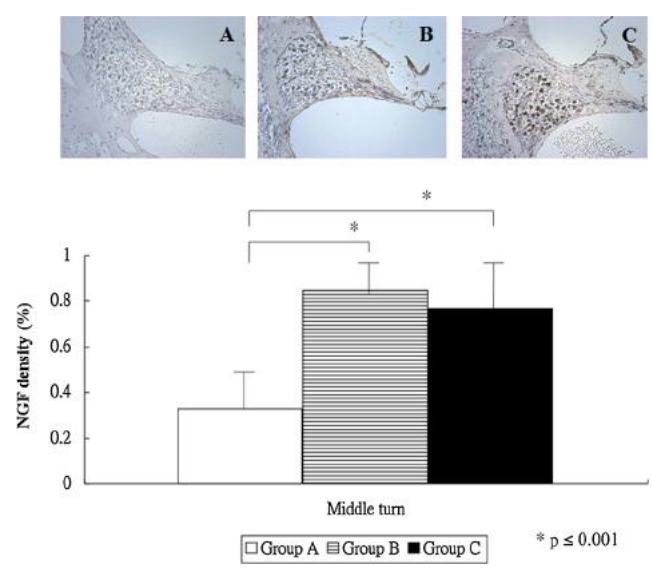

Fig. 1. Results of immunohistochemistric studies for NGF staining at the SGN in the middle turn of the cochlea. NGF density was counted as NGF stained cell number/total cell number (%) under 200X magnification. (n = 8/group) Values were mean ± S.E.M. A = Control, B = 215.25 mg/Kg BW HE, C = 430.50 mg/Kg BW HE.

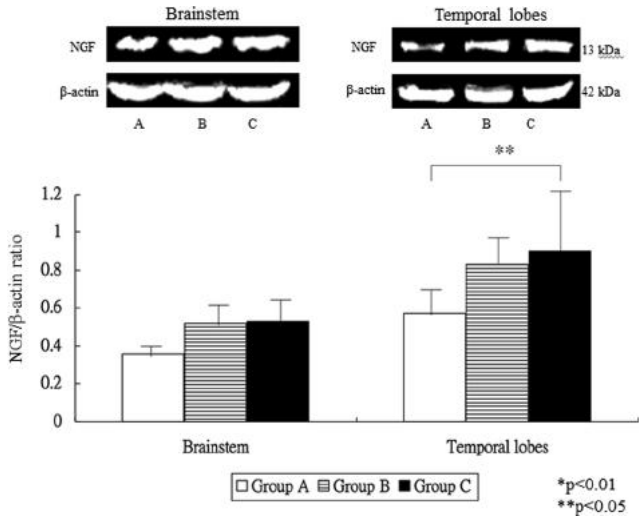

Fig. 2. Electrophoretograms and the ratio of NGF versus β-actin by Western blotting in the three groups at the end of study. The y-axis denotes protein level expression, as determined by the ratio of NGF/β-actin. (n = 8/group) Values were mean ± S.E.M. A = Control, B = 215.25 mg/Kg BW HE, C = 430.50 mg/Kg BW HE.

p = 0.071, respectively). But, group C had significantly higher expressions of mature NGF in the temporal lobes (p = 0.0396; Post-hoc analysis, p = 0.049). However, the BDNF versus α-tubulin expressions did not differ among three groups in brainstem and temporal lobes (data was not shown; p = 0.1973 and 0.1645, respectively).

3.3. SGN density and hair cell numbers

Fig. 3 showed the histology in the middle turn of the cochlea by H&E staining. Compared to group A (6.7 ± 1.18), groups B (8.7 ± 1.43) and C (9.3 ± 1.96) increase the SGN cell density in the middle turn of the cochlea of borderline significance (p = 0.056) (Fig. 3a). Compared to group A (2.4 ± 0.42), group C (3.0 ± 0.00), but not group B (2.8 ± 0.27), had significantly higher OHCs counts in the middle turn of the cochlea (p = 0.019; Post-hoc analysis, p = 0.020 for group C versus group A, respectively). But, the IHCs cell counts in the middle turn of the cochlea was not significantly different in three groups (1.0 ± 0.00 for all groups) (Fig. 3b).

3.4. Expressions of apoptotic factors

Fig. 4 showed the results of immunohistochemistric studies for AIF and PARP-1 in the middle turn of the cochlea. Compared to group A (0.852 ± 0.073), groups B (0.552 ± 0.209) and C (0.082 ± 0.072) had significantly reduced AIF expressions in the SGN of cochlear middle turn (p < 0.0001; Post-hoc analysis, p = 0.001 for group B versus group A; p < 0.001 for group C versus group A and for group C versus group B) (Fig. 4a). Compared to groups A (0.606 ± 0.175) and B (0.486 ± 0.127), group C (0.218 ± 0.087) had significantly lower AIF expressions in the OHCs of cochlear middle turn (p = 0.0020; Post-hoc analysis, p = 0.002 for group C versus group A; p = 0.025 for group C versus group B) (Fig. 4b). In addition, compared to groups A (0.650 ± 0.137) and B (0.450 ± 0.274), group C (0.050 ± 0.112) had significantly decreased AIF expressions in the IHCs of cochlear middle turn (p = 0.0009; Post-hoc analysis, p = 0.001 for group C versus group A; p = 0.017 for group C versus group B) (Fig. 4b). Compared to groups A (0.686 ± 0.203) and B (0.646 ± 0.216), group C (0.216 ± 0.254) had significantly reduced PARP-1 expressions in the SGN of cochlear middle turn (p = 0.0111; Post-hoc analysis, p = 0.019 for group C versus group A and p = 0.032 for group C versus group B) (Fig. 4c). Compared to groups A (0.640 ± 0.037) and B

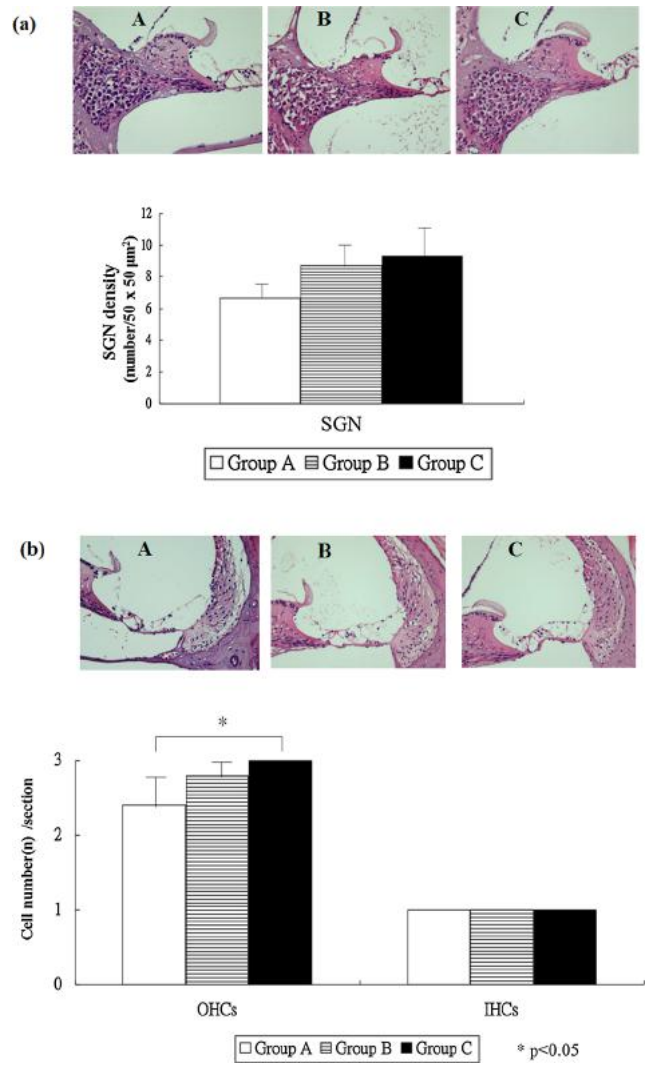

Fig. 3. Histology of the middle turn of the cochlea by H&E staining. (a) SGN cell density (b) OHCs and IHCs cell numbers. SGN cell density was counted as SGN cell number/50x50 μm^2, and OHCs and IHCs cell numbers were counted as stained cell number/section under 200X magnification (n = 8/group). Values were mean ± S.E.M. A = Control, B = 215.25 mg/Kg BW HE, C = 430.50 mg/Kg BW HE.

(0.586 ± 0.067), group C (0.348 ± 0.070) had significantly decreased PARP-1 expressions in the OHCs of cochlear middle turn (p < 0.0001; Post-hoc analysis, p < 0.001 for group C versus group A and for group C versus group B). In addition, groups B (0.500 ± 0.177) and C (0.200 ± 0.274) had significantly declined PARP-1 expressions than group A (0.900 ± 0.137) in the IHCs of cochlear middle turn (p = 0.0006; Post-hoc analysis, p = 0.028 for group B versus group A, and p < 0.001 for group C versus group A) (Fig. 4d).

Fig. 5 showed the ratio of AIF and PARP-1 versus Lamin A by Western blotting in the brainstem and temporal lobes. Compared to group A (1.18 ± 0.17), groups B (0.86 ± 0.22) and C (0.87 ± 0.17) had significantly lower expressions of AIF in the brainstem (p = 0.0159; Post-hoc analysis, p = 0.033 for group B versus group A and p = 0.036 for group C versus group A). Similarly, compared to group A (4.99 ± 1.04), group C (3.13 ± 0.99) had significantly lower expressions of AIF in the temporal lobes (Post-hoc analysis, p = 0.031 for group C versus group A) (Fig. 5a). Compared to group A (0.58 ± 0.10), groups B (0.40 ± 0.09) and C (0.40 ± 0.09) had significantly lower expressions of PARP-1 in the brainstem (p = 0.0034; Post-hoc analysis, p = 0.009 for group B versus group A

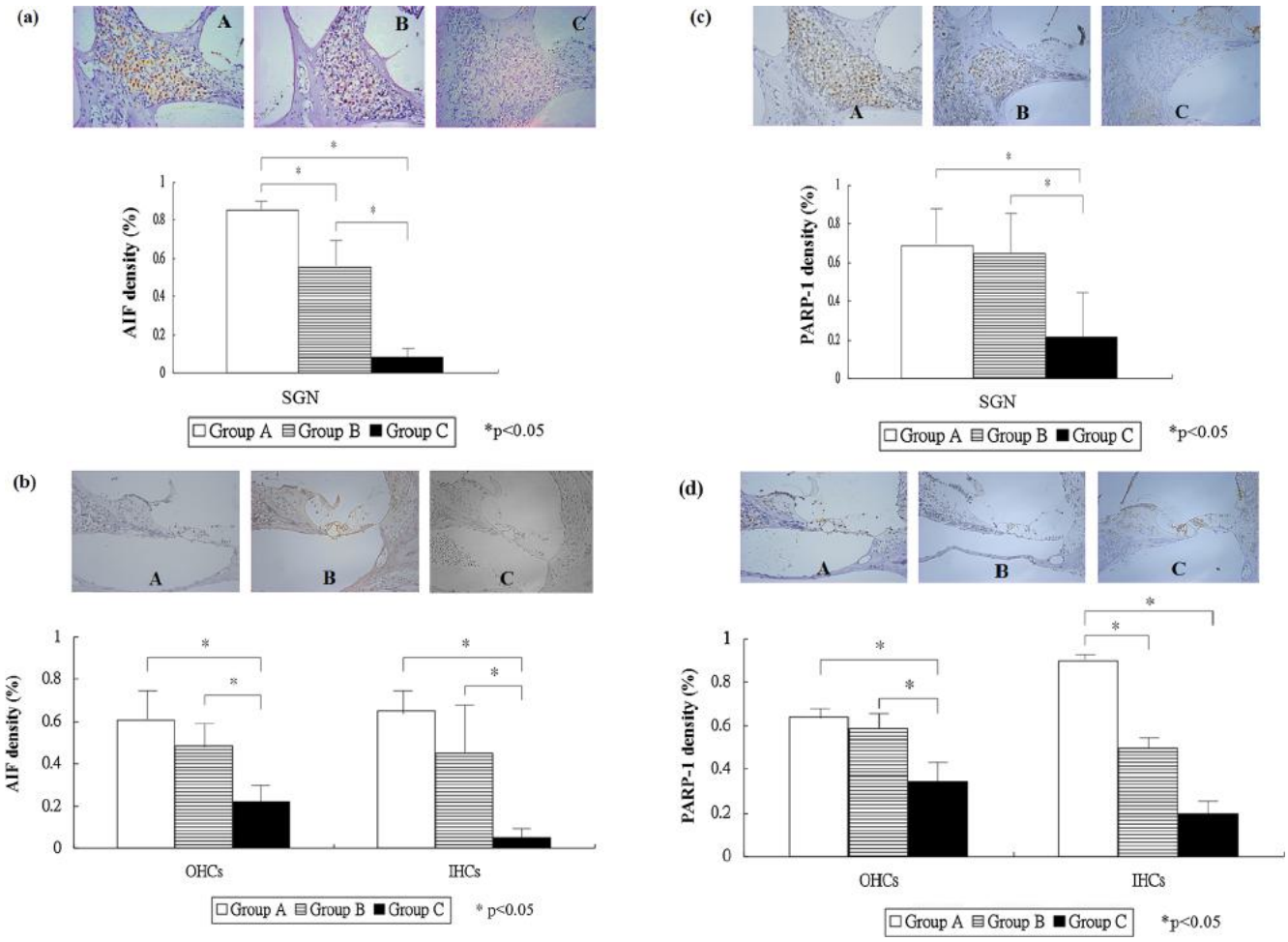

Fig. 4. Results of immunohistochemistric studies for AIF and PARP-1 stainings in the middle turn of the cochlea. (a) AIF density in the SGN. (b) AIF density in the Organ of Corti (c) PARP-1 density in the SGN (d) PARP-1 density in the Organ of Corti. AIF and PARP-1 densities were counted as AIF and PARP-1 stained cell number/total cell number (%) under 200 X magnification, respectively. Values were mean ± S.E.M. (n = 8/group) A = Control, B = 215.25 mg/Kg BW HE, C = 430.50 mg/Kg BW HE.

and p = 0.008 for group C versus group A). Similarly, groups C (1.54 ± 0.46) had significantly lower expressions of PARP-1 in the temporal lobes when compared to group A (2.76 ± 0.79) (p = 0.0133; Post-hoc analysis, p = 0.011 for group C versus group A) (Fig. 5b).

4. Discussion

SAMP8 mice displayed premature hearing loss that mimics human ARHI. Higher oxidative stress, increased neural inflammation, declined endogenous antioxidant activities, and increased apoptosis were observed in the cochlea of SAMP8 mice (Menardo et al., 2012). In this study, our results demonstrate that HE could prevent hearing loss with age in SAMP8 mice possibly via increasing the protein expression of mature NGF along the auditory pathway, and decreasing caspase-independent cell apoptosis in the cochlea (Graphical abstract).

In the healthy cochlea, the survival of SGNs is assured by endogenous neurotrophic support. NGF and BDNF are involved in neuronal survival, dendritic arborization, synaptic plasticity, and neural development. NGF enhances SGNs survival after gentamicin treatment (Kimura et al., 1999), and was involved in postinjury repair and regeneration of the basilar papilla following overexposure to noise (Sliwinska-Kowalska et al., 2005). NGF might reduce the ABR threshold shifts in A/J mice by preserving SGN and OHCs in the cochlea (Gao et al., 2017). The elevated NGF level in SGN cells was associated with hearing improvement in trigonelline-induced hearing loss (Castañeda, Rodriguez, Nam, Hong, & Kang, 2017). In patients with sensorineural hearing defects, the NGF level in serum was relatively lower (Salvinelli et al., 2002). Thus, NGF was suggested to be an otoprotectant for progressive hearing loss (Wang et al., 2017). In details, NGF exerted its roles in sensory hearing impairment of humans through binding with TrkA in the cochleae (Salvinelli et al., 2002). After binding with TrkA, NGF was phosphorylated and then inhibited the apoptosis pathway (Lee et al., 2015).

As for BDNF, it is required for the survival of vestibular ganglion and type II SGNS in the inner ear (Schimmang et al., 2003). The reduction of BDNF signaling in the postnatal cochlea increases the loss of OHC innervation in the cochlea and leads to hearing impairment (Schimmang et al., 2003). BDNF-loaded nanoparticles significantly increase the survival of SGNs in NIH3T3 fibroblasts cell cultures (Schmidt et al., 2018). BDNF also time-dependently prevents noise-induced and cisplatin-induced inner ear damage (Lidian, Stenkvist-Asplund, Linder, Anniko, & Nordang, 2013).

Production and/or use of neurotrophic factors are expensive and, as of yet, cannot be widely employed. Therefore, searching for inexpensive options for hearing protection, is a crucial endeavor. Out of 2000 different types of edible and/or medicinal mushrooms, only a few countable mushrooms might have neuroprotective activities (Samberkar et al., 2015). HE aqueous extract induces NGF synthesis and promotes neurite outgrowth in NG108-15 cells (Lai et al., 2013). HE mycelium increases the ratio of NGF/NGF precursor and hippocampal

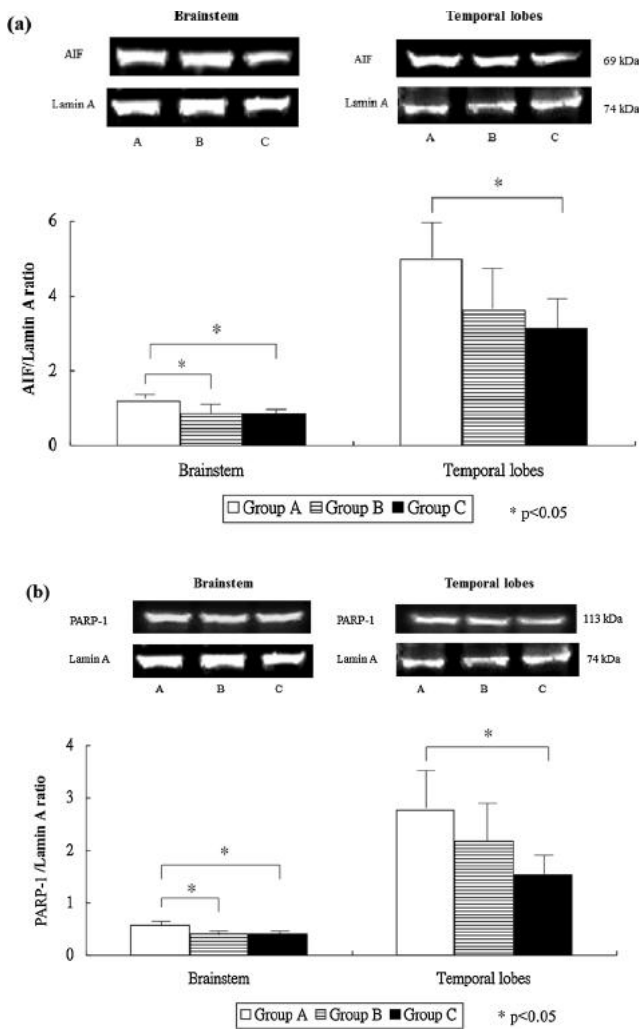

Fig. 5. Electrophoretograms and the ratio of AIF and PARP-1 versus Lamin A by Western blotting in the brainstem and temporal lobes. The y-axis denotes protein level expression, as determined by the ratio of AIF/Lamin A and PARP1/Lamin A in (a) AIF expressions (b) PARP-1 expressions, respectively. (n = 8/group) Values were mean ± S.E.M. A = Control, B = 215.25 mg/Kg BW HE, C = 430.50 mg/Kg BW HE.

neurogenesis (Tzeng et al., 2016). Also, a purified polysaccharide isolated from HE mycelium could prevent H_2O_2-induced neural cell damages by inhibiting apoptotic signaling pathways (Lee et al., 2015), and prevent Aβ-induced PC12 cell damage (Cheng, Tsai, Lien, Lee, & Sheu, 2016). HE triggered more neurite outgrowth in brain, spinal cord, and retinal cells from chick embryo than BDNF did (Samberkar et al., 2015). Our results indicated that HE could increase NGF expression but did not enhance BDNF expression in the cochlea and brain.

On the other hand, AIF and PARP-1 were essential mediators for caspase-independent apoptosis (Delavallee, Cabon, Galan-Malo, Lorenzo, & Susin, 2011). The upregulation of AIF may be important in the process of SGN damage and apoptosis (Ding et al., 2015). Once the AIF was released from mitochondria, it would relocate into the nucleus and induce caspase-independent chromatinolysis (Candé et al., 2004). During this process, overexpression of PARP-1, a nuclear enzyme responsible for DNA damage repairing, would help AIF translocation (Wu, Liu, & Li, 2012). In the auditory system, caspase-independent apoptosis was found in noise-induced hearing loss (Rebillard & Bryant, 1989), SGN cell death (Park et al., 2002) and obesity-induced hearing degeneration (Hwang, 2017).

As for treatment effect, HE mycelium polymer attenuated the apoptosis of PC12 cells (Park et al., 2002). The purified polysaccharide isolated from HE mycelium increased the capacity of oxygen radical absorbance and lessened the H_2O_2-induced apoptotic cell death (Kuo et al., 2017). In this study, the number of IHCs did not change, but AIF and PARP1 percentages decreased significantly in IHCs, as shown in Figs. 3 and 4. We supposed that this discrepancy might be related with caspase-dependent apoptosis pathway which was not assayed in this study. Moreover, we found that HE could prevent ARHI of SAMP8 mice via the axis of NGF and caspase-independent apoptosis signal pathway, although we did not assay the expression of TrkA because of sampling limitations. We hope to investigate the detailed mechanisms linking NGF and caspase-independent apoptosis in the future.

In conclusion, diet supplementation of HE could prevent hearing deterioration in SAMP8 mice. The beneficial effects of HE on hearing might be possibly associated with increasing neurotrophic expressions and decreasing cell death in the cochlea, brainstem and temporal lobes. These experimental results might encourage the proposals of clinical trials in the future.

Ethics statement

I have read and adhere to the Publishing Ethics.

Author contributions

YC Chan and JH Hwang conceptualized the experiments. HT Chiang performed the experiments. HT Chiang, CC Chen, LY Lee and MF Wang curated the experimental data. YC Chan and JH Hwang wrote original draft. YC Chan wrote - review & editing. All authors read and approved the final manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jff.2020.103832.

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